AgenticRotation: Supplementary Material

Anonymous CVPR submission

A. Scope of This Release

This supplement accompanies the working manuscript updated on 1 November 2026. All measured results concern the marked RSLCube. The original release contains 400 trials, followed by separate replication, intervention, and controller-development cohorts. The latest completion bundle contains 760 live semantic trials and a separate 160-trial fixed-plan development test. Its 240 bare-VLM visual trials overlap the earlier visual archive and are not new trials. Historical outcomes are retained. Missing multi-object, improved-predictor, and physical-hand measurements are marked TODO.

The latest complete-system comparison refits only Astra's waiting actor. The student, predictor, execution actor, and task criterion remain frozen. Earlier cohorts use their recorded original controller version. Results from the earlier RGB-D backend, invalid contact-capacity runs, and different protocols are not pooled into the current main result. The original bare-baseline addendum was specified after the first suite, so the full historical study is not described as blindly preregistered.

B. Exact Task and Runtime Contract

B.1. Object and goals

Table 1 records the fixed face mapping. Body normals include the original mesh rotation. All methods use the same mapping and cached six-vector bank. The cyan channel corresponds visually to a pale-blue face and magenta to a purple face. The target quaternion gives a canonical twist as well as the upward face normal.

Letter	Skill color	Body normal	Target XYZW
T	red	$(0, -1, 0)$	$(-s, 0, 0, s)$
E	green	$(-1, 0, 0)$	$(0, s, 0, s)$
R	blue	$(0, 0, -1)$	$(1, 0, 0, 0)$
L	yellow	$(0, 0, 1)$	$(0, 0, 0, 1)$
S	magenta	$(0, 1, 0)$	$(s, 0, 0, s)$
H	cyan	$(1, 0, 0)$	$(0, -s, 0, s)$

Table 1. Frozen goal mapping, with $s = \sqrt{1/2}$.

The semantic instruction identifies a letter. It does not specify an arbitrary target quaternion. The benchmark scores its associated canonical quaternion at a 0.16-radian tolerance. A drop is recorded when the distance from the object's nominal grasp position is at least 0.24 metres. This is an operational benchmark termination, not a detector of every transient slip or lost contact. The object may move within that boundary without a recorded drop.

B.2. Physics and perception

The runtime uses Isaac Lab 3.0, Isaac Sim 6.2, Newton 1.6.1, Warp 1.17, and PyTorch 2.11 with CUDA 13.0. The validated collision model contains 23 bodies and 117 shapes, comprising 116 convex shapes and one box. The contact allocation uses `nconmax=4096` and `njmax=1024`. Earlier overflow runs are invalid even if their reported rewards appear plausible.

The pose estimator is frozen. It predicts 27 image-coordinate/depth outputs for nine points, then recovers pose through calibrated geometry. The object image is rendered at 120×120 with live hand occlusion. Sensor calibration, object asset, policy normalization, joint order, action mapping, and smoothing belong to the release contract. A change to one of these produces a new release. The current setup does not establish real-camera robustness or unseen-scene generalization.

B.3. Timing, termination, and exclusions

The semantic horizon is 120 seconds of logical control time. Model latency is measured independently and translated into a minimum pending duration of $\lceil 30L \rceil$ steps. The request must also have actually completed. Physics is never intentionally paused to await a semantic response. Because simulation throughput is below the logical 30 Hz rate, wall time per episode exceeds logical duration. A hardware claim requires a separately measured end-to-end rate.

The shared gate runs at 3 Hz. It requires an accepted requested-symbol binding, a non-inspection command, no outstanding request, and estimated canonical orientation error no greater than 0.16 radians. The independent scorer applies the same angular tolerance to true orientation and checks the actual requested face. No post-stop dwell is required. Success is not the minimum orientation error at any earlier time.

A service error or model call exceeding 120 seconds is recorded without replacement. The original three-system panel removes trial 183102011 from every arm. The four-way control panel removes 183102000, 183102010, and 183102011. The resulting denominators are 39 and 37. Their percentages are not directly interchangeable. Raw 40-trial outcomes remain visible, including service-affected cases.

C. Models and Training

C.1. Semantic and motor inputs

The motor policy receives 109 target-free proprioceptive/history features, seven measured pose values, and a 768-dimensional goal. Its goal projection has width 64 and a tanh nonlinearity. The inherited recurrent policy emits 21 normalized joint targets. The asset-specific goal quaternion is absent from the migrated student's inference inputs.

The hosted model route recorded in the manifests is `gpt-6-astra`. The local grounding backbone is Qwen3-VL-4B-Instruct. The Astra pathway uses public tool actions and a deterministic goal-bank lookup. The fine-tuned pathway predicts a continuous vector within the normalized skill span. Neither pathway receives the requested face's true pose. The local unreadable-view schedule is $(1, 3, 0, 2, 4, 5)$ with 90-step inspections. Astra chooses inspection actions of 10–90 steps. This scheduler difference is part of the complete-system comparison and a confound for attributing all effects to model weights.

[TODO: Document the hosted model's exact serving revision, sampling settings, and full sanitized system prompt/tool schema in the final anonymous release. A route name alone does not guarantee reproducibility across provider updates.]

C.2. Grounding supervision

The grounding model first uses 4,096 teacher-view training examples and 1,024 validation examples, then 4,800 student-view training examples and 2,400 validation examples with mixed replay. Each adaptation stage uses 240 updates. Training and validation collection seeds are disjoint from the formal evaluation seeds. Readability is a geometric/image-signal proxy. The objective combines readability prediction, readable-example skill classification, and cosine alignment to a canonical vector. We do not claim human-verified text recognition labels or an action-consistency objective in this release.

The fixed layout permits learned letter-to-color shortcuts. A held-out pose or trajectory alone does not remove that possibility. A convincing grounding test changes or hides the visual evidence while holding the instruction and physical state fixed, then checks whether the model's selected target changes appropriately.

C.3. Controller features and losses

The sensor branch uses the current proprioceptive vector, measured pose, student belief and proposal, request-anchor pose and action, fitting residual, binding flag, pending flag, and elapsed pending duration. Its normalized inputs are clipped to $[-8, 8]$. Goal features are centered and scaled using the fixed bank. The sensor, goal, and fusion widths are 192, 64, and 128. Each actor has 22 outputs: a speed logit and 21 residual values.

The waiting actor interpolates from current joint positions toward the proposal and adds a bounded residual. The execution actor applies a bounded residual to the proposal at full speed. Both are supervised from the teacher with residual regularization of 0.01. Waiting labels hold the request anchor. Execution labels follow the active goal. Paired state/goal examples and schedule-derived states are used during fitting. Their causal utility remains to be established by matched training ablations.

No raw RGB, true object pose, teacher action, future model response, or oracle semantic label is a deployed actor input. Vision affects the actor through estimated pose and recurrent belief. The reported stopping gate is measured-pose based. Previously trained completion-evidence heads are inactive and cannot be credited for the current results.

D. Original-Release Numerical Results

Semantic condition	Seed 1	Seed 2	Raw total
Astra full	13/20	16/20	29/40
Fine-tuned VLM, bare	14/20	13/20	27/40
Rule VLM, bare	5/20	9/20	14/40
Fine-tuned VLM, shared control	15/20	15/20	30/40
Rule VLM, shared control	6/20	10/20	16/40
Astra, fixed slow / student	11/20	12/20	23/40
Astra, learned wait / student	15/20	11/20	26/40
Astra, fixed slow / learned exec.	16/20	12/20	28/40

Table 2. Raw semantic successes by seed. Seed 1 is 183101 and seed 2 is 183102. The bare addendum contributes the second and third rows only.

Comparison on eligible trials	A-only	B-only	Exact p
Full Astra / bare fine-tuned	9	6	0.6072
Full Astra / bare rule	18	2	0.0004
Full Astra / fixed control	15	8	0.2100
Full Astra / waiting only	8	6	0.7905
Full Astra / execution only	8	7	1.0000

Table 3. Exploratory two-sided exact McNemar tests. The first two rows use 39 trials and the last three use 37. Tests are unadjusted and do not establish all module effects.

On raw paired trials, adding shared control to the fine-tuned VLM changes successes from 27 to 30 and drops from nine to five. Five trials succeed only without the actors and eight only with them, giving $p = 0.581$. The rule VLM changes from 14 to 16 successes, with $p = 0.754$. These auxiliary results must remain available even though the primary baselines omit the actors. They show that controller access can affect a complete-system comparison and that current evidence does not establish an Astra-specific benefit.

Individual rate intervals use the Wilson method with nominal 95% coverage. Paired comparisons use trial identities and initial-state verification. They account for pairing but do not address all uncertainty from hosted model variability or future object distributions. A single two-seed cohort is insufficient to establish a stable ordering between systems separated by two or three raw successes.

E. Successor Student: A Separate Development Study

The existing cube results use a fixed student checksum. Successor training is designed to ask whether extra DAgger with the actual frozen CNN sensor improves physical control beyond an equal budget with simulated pose noise. Both branches start from that same latent student, use training seed 183201, and run 400 updates with 16 environments, 45 rollout steps, and four fitting epochs. Learning rate is 3×10^{-4} with a new Adam state and zero weight decay. The teacher action fraction decays as $0.9 \cdot 0.95^{i}$. Each branch collects 288,000 transitions. The CNN and teacher remain frozen.

After training, the final checkpoint from each branch is compared with the released student and the frozen latent student. Every arm uses CNN observations and 40 fresh diagnostic trials from seeds 183221 and 183222. No intermediate checkpoint is selected from those trials. The CNN branch must exceed both latent references in reach count without increasing drops to pass the development admission rule. This gate is an engineering selection rule, not a significance claim.

A selected new student still requires separately collected controller data and compatibility validation. Only then may a new semantic release use reserved seeds 183301 and 183302. The original cube-v1 results remain unchanged. Both branches and all 160 diagnostic trials have completed with valid physics logs. The four reach counts are 39/40, 38/40, 39/40, and 40/40 for the released, frozen latent, extra simulated, and extra CNN policies. Drops are one, two, one, and zero. The CNN candidate passes the stated gate and has not been automatically deployed. Exact paired tests against the frozen latent and extra simulated policies give $p = 0.5$ and $p = 1.0$. Thus this is a promising development result, not an established perception-matching effect.

F. Separate Follow-up Cohorts

These studies retain the frozen task success criterion and do not replace the completed cube-v1 cohort. Service-eligible success is the primary metric. Cross-method comparisons remove the same service-affected slots from every arm. All raw records, including excluded physical drops, remain available. Forty scheduled slots do not necessarily yield forty eligible trials.

F.1. Completed independent replication

Seeds 183411 and 183412 each provide 20 trials per method. All 120 scheduled records pass the source, checkpoint, physical-log and initial-state audits. Slots 183411008, 183411015 and 183411017 have recorded Astra service errors and are removed from every arm for the matched comparison. The last includes a connection failure after approximately 126 seconds and a waiting-phase drop. Its exclusion follows the existing service rule. Raw success counts are 26/40, 29/40 and 22/40 for Astra, bare fine-tuned VLM and bare rule VLM. The common eligible counts appear below.

Method	Eligible success	Rate (%)
Astra full	26/37	70.3
Fine-tuned VLM, bare	27/37	73.0
Rule VLM, bare	21/37	56.8

Table 4. Completed independent replication on 37 common service-eligible initial states from 40 scheduled trials per method. This supersedes the archived partial snapshot and remains separate from the original cube-v1 cohort.

The eleven eligible Astra failures comprise nine drops, one false stop and one abstention. The drops include one during startup, two during subsequent waiting and six outside waiting. Raw Astra counts additionally include one service-excluded drop. Against the fine-tuned VLM, Astra has five exclusive successes and six exclusive failures (exact paired $p = 1.0$). Against the rule VLM, the counts are eleven and six ($p = 0.332$). These exploratory comparisons do not establish superiority. The earlier 18-slot snapshot remains archived for provenance but is no longer used for the follow-up table.

F.2. Control interaction and explicit goals

The historical factorial uses a common 37-trial panel, with successes of 20, 25, 26 and 27 for fixed control, waiting only, execution only and both actors. Learned waiting alone reduces startup drops from four to one but introduces three later waiting drops, leaving total waiting-related drops at four. With execution recovery present, learned waiting eliminates four startup drops while execution drops increase from four to nine. The full method wins eight trials lost

by execution-only control and loses seven won by that control. Hence net success improves by one, without a resolved interaction benefit.

A deterministic restoration diagnostic verifies 19 of 20 planned physical states. Each state has two assigned goals and both actor conditions, yielding 76 valid branches. State 12 cannot be restored because its prefix terminates, and is not replaced. Both conditions give the assigned goal to the student. No waiting drops occur, but their subsequent execution differs.

Actor input	Final in tolerance	Drops
Explicit actor goal	26/38	5
No explicit actor goal	29/38	2

Table 5. Paired recovery diagnostic on 19 states, two goals each. Four branches in each condition already start within tolerance. Final tolerance is not autonomous semantic success. Explicit actor-goal conditioning is not supported as beneficial by this diagnostic.

Mean assigned-goal progress is 1.195 radians with explicit actor-goal input and 1.340 without it. Different first actions under different goals do not isolate the actor's explicit goal pathway because the student itself is goal-conditioned. The completed semantic comparison gives 25/36, 24/36 and 30/36 successes for correct, no-explicit and shuffled controller goals. Four service-affected slots are excluded symmetrically. This does not establish a benefit from the explicit goal input.

F.3. Completed visual interventions

The upper-level image changes while the low-level pose predictor continues to receive the original image. Lighting is fixed. The permuted condition assigns the next symbol in (T, E, R, L, S, H) to each color face and changes the requested symbol accordingly, preserving the scored physical target across conditions. The masked condition hides the lettering. Geometry-consistent image construction and the transmitted image hashes are audited. Seeds 183441 and 183442 provide 20 trials each, and physical initial-state hashes match across the three conditions.

Method	Identity	Permuted	Masked
Astra full	32/40	32/40	0/39
Fine-tuned VLM, bare	30/40	16/40	3/40
Rule VLM, bare	23/40	23/40	0/40

Table 6. Completed upper-image interventions. Identity and permutation each use 40 trials. Astra masking has one service exclusion. For its paired masking contrast the identity numerator becomes 31/39. All conditions use reconstructed face images, not the untouched native-image comparison.

Fine-tuned success falls by 35 percentage points after remapping and to 7.5% after masking. Rule-based success is 57.5% under both identity and remapping, then zero after masking. Equal aggregate identity and remapped rates do not imply identical trial outcomes. Astra now has 32/40 successes under both identity and permutation, with zero of 39 eligible masked trials. This supports robustness to the tested permutation, without proving equivalence or identifying all fine-tuned successes as a fixed lookup.

Applied semantic bindings. We audit every applied, accepted call from the same 240 trials. The episode-file hashes match the archived results, and the last accepted skill agrees with the recorded terminal binding. Table 7 separates acquiring the correct target from completing it. Under permutation, the fine-tuned model first binds correctly in 25 trials and incorrectly in eleven, compared with 34 and four under identity. Nine of its eleven wrong-binding trials select the original color for the newly requested symbol. The rule model has six wrong-binding trials under each layout, with three following the old association after permutation. This pattern is consistent with reliance on training associations, but does not identify the model's internal mechanism.

Model	Layout	Correct	Wrong	None	Old
Fine-tuned	Identity	34	4	2	–
Fine-tuned	Permuted	25	11	4	9
Fine-tuned	Masked	3	1	36	–
Rule	Identity	24	6	10	–
Rule	Permuted	25	6	9	3
Rule	Masked	0	0	40	–

Table 7. First accepted binding in each of 40 trials. Old counts trials with an accepted skill matching the original symbol–color association, defined only for permutation. These are the same trials as the success table, not new evaluations.

No wrong-to-correct binding transition occurs in these bare-VLM cohorts. This is descriptive: trajectories, visibility, and opportunities for additional calls differ. Of fourteen fine-tuned false stops under permutation, ten end with a wrong binding and four with the correct binding but an independently incorrect final pose. Masking leads to no accepted binding in 36 fine-tuned trials and all forty rule-model trials. The three masked fine-tuned successes cannot establish letter grounding when that information is removed.

Identity versus permutation yields eighteen exclusive successes and four exclusive failures for the fine-tuned model (exact paired $p = 0.00434$), versus ten and ten for the rule model ($p = 1.0$). The final analysis applies Holm correction across six planned image contrasts, giving adjusted $p = 0.0130$ for the fine-tuned mapping contrast. The single deterministic remapping and reconstructed upper im-

ages limit generalization claims. Astra has five exclusive identity successes and five exclusive permutation successes ($p = 1.0$).

F.4. Completed feedback comparison

The same-Astra comparison uses one initial bounded plan versus at most six feedback calls, with the original actors and frozen student, stopping gate, and horizon. Seeds 183431 and 183432 give 40 trials per arm. Slot 183432001 is service-affected and excluded from both, leaving 7/39 versus 30/39 successes. Raw counts are 7/40 and 31/40. Actual model calls total 40 and 113. Twenty-four trials succeed only with feedback and one only with the initial plan ($p = 1.55 \times 10^{-6}$). Seventeen initial-plan abstentions become feedback successes. No wrong-to-correct binding transition is observed. Unequal inference budgets prevent isolating a pure feedback effect at equal compute.

F.5. Latest native comparison and failure audit

Seeds 185451 and 185452 give 40 matched trials per method with no service exclusions. The full Astra system gives 35 successes, three execution-phase drops, one wrong-binding false stop, and one abstention. The bare fine-tuned system gives 30 successes, nine execution drops, and one horizon failure. The bare rule system gives 17 successes, twelve drops, ten false stops, and one horizon failure. Its drops include two during startup waiting. Astra makes 101 calls, fine-tuning 117, and the rule baseline 126. Median latencies are 4.349, 0.099 and 1.104 seconds.

The three Astra drops occur during inspection before final target binding. One is 59 control steps after the most recent applied inspection command, and the others are 36 and 24 steps after it. These intervals describe timing, not causal attribution. The false-stop episode binds R to cyan rather than blue. Its measured error to the declared target is 0.057 radians, while true requested-target error is 3.079 radians. It is a semantic binding failure and cannot simply be assigned to pose estimation. The abstention follows four inspections without a declared target. A favorable instantaneous pose during that trajectory does not meet the autonomous success rule.

F.6. Waiting refit and live confirmation

The refit uses training collection seed 185301, validation seed 185302, and fitting seed 185311. The inherited actor is updated 2,400 times. Actor batches allocate one half to pending states and one quarter each to unbound and bound execution states. Pending samples are uniform. A startup-balanced alternative replaces that half-batch with equal early and later pending samples, holding the other training settings fixed. Only final-iteration candidates are evaluated. Separate fixed-plan development seeds 185321 and 185322 select the uniform candidate under a preset completion/drop gate. Only its waiting outputs replace the deployed actor. Execution remains unchanged.

Live confirmation uses seeds 185341 and 185342. Slot 185341008 has a refit service failure and is removed symmetrically. Original waiting gives 27/39 successes and ten drops, compared with 33/39 and five drops after refitting. Raw success is 28/40 versus 33/40. There are eight refit-only and two original-only successes ($p = 0.109375$). The drop counts also apply to the eligible panel. Startup, later-waiting and execution drops change from $(2, 2, 6)$ to $(0, 1, 4)$. These data are distinct from the latest native three-system cohort.

F.7. Goal-input intervention

Seeds 183421 and 183422 provide 40 trials per condition. Correct, retrained no-explicit-goal and shuffled controller goals give raw successes of 29/40, 26/40 and 31/40. Four symmetrically excluded slots leave the following panel. The student still receives its assigned goal, so removing the explicit actor-goal vector does not remove all goal information from the controller's proposal and belief inputs.

Actor input	Eligible success	Rate (%)
Correct goal	25/36	69.4
No explicit goal	24/36	66.7
Shuffled goal	30/36	83.3

Table 8. Completed goal-input intervention. No positive explicit-goal benefit is established. Shuffled goals are not selected as a new method after observing these results.

F.8. Rejected execution-gate development

This separate 160-trial study replays fixed public decisions and delays, with no new Astra calls. Every arm uses the same refitted waiting actor. During non-pending steps, a gate interpolates between the student's nominal proposal and the original execution actor's corrected action. Waiting actions remain unchanged. Seeds 185481 and 185482 provide 40 trials per arm. The fixed gate is 0.8282. The phase-fixed gate uses 0.5872 when unbound and 0.9877 when bound. The learned candidate must exceed original execution by at least three successes, exceed the best fixed gate

Execution condition	Success	Drops
Original execution	29/40	3
Fixed residual gate	27/40	7
Phase-fixed gate	24/40	10
Learned residual gate	26/40	5

Table 9. Fixed-plan development outcomes, not live Astra evaluation. The learned gate is rejected and absent from the latest main system.

by at least two, and introduce no more drops than the best control. It achieves 26 successes against required counts of 32 and 29, with five drops against a limit of three. All gates fail. Both per-seed outcomes and all negative results are retained. No additional live evaluation is triggered by this candidate.

F.9. Timing verification

The latest native suite contains 344 actual calls, of which 342 are applied before episode termination. The remaining two replies belong to terminated rule-baseline episodes and are retained. The timing audit checks 52,475 control steps. Separate audits cover 80 waiting trials, 80 feedback trials and 360 visual trials. Pending requests preserve continuous logical control, and replies are applied only after their measured latency and actual completion. These checks verify the delay protocol rather than wall-clock real-time operation.

G. Experimental Requirements Before Submission

G.1. Separate model strength from control

The primary comparison uses full Astra against bare VLM baselines. Its interpretation is system-level. To attribute gains to the upper-level model, compare all providers with identical control actors, timing conventions, stopping, and information access. Preserve the existing shared-control results. Repeated live calls can produce different public plans even on matching initial states. Fixed-public-plan replay can isolate motor changes, but replay success must not be reported as a new live Astra success rate.

G.2. Test semantic and visual causality

The completed goal-input and restored-state tests do not establish a useful explicit controller-goal pathway. The completed visual intervention supports sensitivity to visible semantics, with only one reconstructed-image permutation. Further claims require additional mappings or objects, and an optional text-only comparison. These extensions must use new declared trials rather than select models on the completed test set.

G.3. Test agent feedback

The completed initial-plan comparison supports the full feedback protocol. An equal-call-budget or history/verification ablation would be needed to isolate those components. Keep the registered one-call and at-most-six-call budgets explicit and record actual calls, logical duration, wall time, and service failures. This contrasts complete protocols and does not isolate fresh information from extra inference compute. Latency replay should use the same public plan and delay sequence across motor variants. A simple time-aware residual actor provides a stronger learning baseline than fixed slowing alone. It should not be labeled an official reproduction of A2C2 or ARLI without a verified implementation.

G.4. Separate perception accuracy and control benefit

Compare the original and improved predictors with the same student and agent first. Report independent translation/rotation error, tail error, keypoint visibility, false stops, and semantic success. Oracle pose estimates the residual motor gap. Only a second experiment should adapt the student to the new predictor. A lower pose error is useful evidence, but it does not guarantee a higher task success rate.

G.5. Objects and physical hand

Select about five objects with variation in geometry, inertia, contact affordances, and semantic ambiguity. Record object-specific students and training budgets. Balance the task set within each object and report each row before computing an object macro-average. Forty trials per arm per object is the default batch size. Hardware runs need calibrated independent scoring, documented resets, two sessions, and synchronized video and command logs. A new object checkpoint alone is insufficient without its assets, calibration, normalization, action conventions, and goal definitions.

H. Reproducibility and Evidence Audit

The archive retains 400 original-release trials, 160 successor-development trials, 120 earlier replication trials, and the 19-state recovery diagnostic. The latest snapshot has 760 live records: native 120, goal 120, feedback 80, visual 360, and waiting 80. Its 240 bare visual records overlap the earlier archive. Another 160 fixed-plan development trials complete the 920-trial bundle. These protocols must not be pooled into one rate. Partial snapshots add no trials.

Sanitized records retain initial-state hashes, goals, measured stops, independent errors, outcomes, exclusions and latencies. Source hashes link to the full archive. Table generation checks counts, matching, exclusions and paired tests. New students, predictors, objects or stop rules require new releases.

[TODO: Complete the anonymous implementation package and object licenses.] The manuscript contains no identifying repository link.